

Started on	Saturday, 8 August 2026, 9:33 AM
State	Finished
Completed on	Tuesday, 18 August 2026, 9:49 AM
Time taken	10 days
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1

Correct

Mark 1.00 out of 1.00

Write the algorithm for [QR decomposition](#) using the Gram-Schmidt method.

For example:

Input	Result
$\begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$	The Q Matrix is $\begin{bmatrix} 0.70710678 & 0.40824829 & -0.57735027 \\ 0.70710678 & -0.40824829 & 0.57735027 \\ 0. & 0.81649658 & 0.57735027 \end{bmatrix}$ The R Matrix is $\begin{bmatrix} 1.41421356 & 0.70710678 & 0.70710678 \\ 0. & 1.22474487 & 0.40824829 \\ 0. & 0. & 1.15470054 \end{bmatrix}$

Answer: (penalty regime: 0. %)

Reset answer

```

1  '''
2  Program to QR decomposition using the Gram-Schmidt method
3  Developed by: VEDHA M
4  RegisterNumber: 212225230292
5  '''
6  import os
7  os.environ["OPENBLAS_NUM_THREADS"]="1"
8  import numpy as np
9  def QR_Decomposition(A):
10     n,m = A.shape
11     q = np.empty((n,n))
12     u = np.empty((n,n))
13     u[:,0] = A[:,0]
14     q[:,0] = u[:,0]/np.linalg.norm(u[:,0])
15     for i in range(1,n):
16         u[:,i] = A[:,i]
17         for j in range(i):
18             u[:,i]-=(A[:,i]@q[:,j]*q[:,j])
19         q[:,i] = u[:,i]/np.linalg.norm(u[:,i])
20     r = np.zeros((n,m))
21     for i in range(n):
22         for j in range(i,m):
23             r[i,j] = A[:,j]@q[:,i]
24     print("The Q Matrix is\n",q)
25     print("The R Matrix is\n",r)
26 x = eval(input())
27 a = np.array(x)
28 QR_Decomposition(a)
29
30

```

	Input	Expected	Got	
✓	(((1, 1, 0], [1,0,1], [0, 1, 1]))	The Q Matrix is <pre>[[0.70710678 0.40824829 -0.57735027] [0.70710678 -0.40824829 0.57735027] [0. 0.81649658 0.57735027]]</pre> The R Matrix is <pre>[[1.41421356 0.70710678 0.70710678] [0. 1.22474487 0.40824829] [0. 0. 1.15470054]]</pre>	The Q Matrix is <pre>[[0.70710678 0.40824829 -0.57735027] [0.70710678 -0.40824829 0.57735027] [0. 0.81649658 0.57735027]]</pre> The R Matrix is <pre>[[1.41421356 0.70710678 0.70710678] [0. 1.22474487 0.40824829] [0. 0. 1.15470054]]</pre>	✓
✓	(((12, -51, 4], [6, 167, -68], [-4, 24, -41]))	The Q Matrix is <pre>[[0.85714286 -0.39428571 -0.33142857] [0.42857143 0.90285714 0.03428571] [-0.28571429 0.17142857 -0.94285714]]</pre> The R Matrix is <pre>[[14. 21. -14.] [0. 175. -70.] [0. 0. 35.]]</pre>	The Q Matrix is <pre>[[0.85714286 -0.39428571 -0.33142857] [0.42857143 0.90285714 0.03428571] [-0.28571429 0.17142857 -0.94285714]]</pre> The R Matrix is <pre>[[14. 21. -14.] [0. 175. -70.] [0. 0. 35.]]</pre>	✓

Passed all tests! ✓

► Show/hide question author's solution (Python3)

Correct

Marks for this submission: 1.00/1.00.